MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY, CHARGE STORAGE

TYPE 8832

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Direct-view charge storage tube, 5-inch flat face, 2-writing guns, electrostatic focus and deflection, aluminized, P20 phosphor, factory collimated, high resolution and high contrast suitable for conventional radar presentation and television display.

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2

ABSOLUTE-MAXIMUM RATINGS: Writing guns - each gun (voltages with respect to writing gun cathode).

Parameter:	Ef	Ehk	Ec1	ec1	Ec3	Ec4	tk
Unit:	V	Vdc	Vdc	vdc (Notes 3 and 5)	Vdc	Vdc	sec
Maximum:	6.93	+125	0	0	2,000	2,750	---
Minimum:	5.67	-125	-200	---	---	---	30
TEST CONDITIONS:	6.3	---	Adjust	---	Adjust	2,490 to 2,510	---

ABSOLUTE-MAXIMUM RATINGS: Writing guns - each gun (voltages with respect to reference input electrode).

Parameter:	ed	Rg	Zd	Ek	Alt
Unit:	v (Notes 2 and 20)	Meg	Meg	Vdc	ft
Maximum:	500	1	0.1	---	60,000
Minimum:	---	---	---	-2,650	---
TEST CONDITIONS:	---	---	---	-2,400	---

ABSOLUTE-MAXIMUM RATINGS: Viewing gun (voltages with respect to reference input electrode). (See note 41.)

Parameter:	Ef	tk	Ebe	Esuppr	Ec2	Escn	Ehk	R Screen	Rbe
Unit:	V	sec	Vdc	Vdc	Vdc	Vdc	Vdc	Meg (Note 4)	Ohms (Note 6)
Maximum:	6.93	---	35	150	175	11,000	+125	---	5 K
Minimum:	5.67	30	-20	-50	---	---	-125	1	---
TEST CONDITIONS:	6.3	---	+15	+75	100	10,000	---	1	---

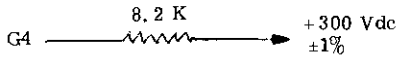
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ABSOLUTE-MAXIMUM RATINGS: Viewing gun - factory collimated section (voltages with respect to reference input electrode). (See note 41.)

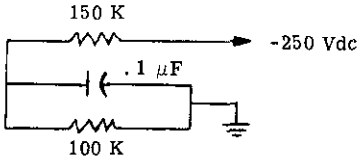
Parameter:	Positive input	Negative input
Unit:	Vdc	Vdc
Maximum:	270	-60
Minimum:	---	-100

TEST CONDITIONS:

Positive input. To be supplied from the following circuit:



Negative input. To be supplied from the following circuit:



GENERAL:

Qualification - Required

Aging - See note 35

Rough-handling test end points - See notes 7, 8, and 9

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification					
1006	Salt spray		---	---	---	---
---	Operation	See note 28	---	---	---	---
	Quality conformance inspection, part 1					
1301	Heater current (each gun)		If	0.5	0.7	A
5201	Electrode current (focusing) (each gun)		Ic	-1	+1	μA
5201	Electrode current (cathode)					
	Radar (writing gun)	See notes 12 and 13	Ik	---	1,000	μA
	TV (writing gun)		Ik	---	1,500	μA
5201	Electrode current (cathode) (viewing gun)		Ik	---	5	mA
5401	Voltage breakdown		---	---	---	---
5401	Voltage breakdown (electrostatic types)	See note 14	---	---	---	---
---	Storage time	See note 15	---	15	---	Sec
5101	Base alinement	3D4, base key	---	---	---	---
5409	Blemishes	See notes 16, 17, and 43	---	---	---	---
---	Loose particles	See note 38	---	---	---	---
5221	Light output					
	Radar	See note 36	---	1,800	5,000	ftL
	TV	See note 44	---	750	1,500	ftL

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued					
---	Erasing					
	Radar	See notes 18 and 19	---	0.002	0.006	Du
	TV	See note 44	---	18	27	%
---	Resolution					
	Radar	See note 21	---	70	---	Lines/in.
	TV	See note 42	---	400	---	TV lines/ 3-1/2 in. line length
5402	Grid-cutoff voltage (writing gun)		Eco	-100	-60	V
5246	Focusing voltage at cutoff		---	210	410	V
5246	Focusing voltage at modulation condition	See note 24	---	75	425	V
---	Suppressor cutoff	See note 19	Eco supp	-5	0	V
5407	Heater-cathode leakage	Ehk = ±100 Vdc	---	---	10	μAdc
5407	Electrode leakage (grid No. 1)		---	---	1	μAdc
5407	Electrode leakage (focusing) (grid No. 3 of writing guns)		---	---	1	μAdc
---	Insulation of electrodes	See note 26	---	100	---	Meg
---	Screen current	Saturated light output	---	---	700	μA
---	Heater-cathode warmup time	Stable operation (see note 39)	---	---	60	Sec
---	Grid drive requirements (writing guns)	See notes 27 and 44; figure 3	---	20	---	%
---	Writing uniformity	See notes 27 and 42; figure 3	---	---	2	Ratio
---	Useful scan		Dia	3.5	---	Inches
---	Vibration	See note 31; figure 4	---	---	1	mm
---	Single pulse erase time	See note 34	---	---	7	ms

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 (see note 40)							
1141	Pressure (implosion)	See note 8	6.5	S3	---	---	---	---
5101	Angle between traces		1.0	II	---	89	91	Degrees
---	Gray scales							
	Radar		1.0	II	---	3	---	---
	TV	See note 42	1.0	II	---	9	---	---
5231	Spot displacement	See note 22	1.0	II	---	---	1	mm

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 (see note 40) -Continued							
5404	Spot position		1.0	II	---	---	15	mm
5406	Deflection factor	1D2	1.0	II	---	82	100	V/in./ 2.5 kV
5406	Deflection factor	3D4	1.0	II	---	82	100	V/in./ 2.5 kV
5406	Deflection factor uniformity		1.0	II	---	---	7	%
---	Pattern distortion	See note 25	1.0	II	---	---	---	---
---	Useful viewing diameter	Concentric with shield	1.0	II	---	4	---	Inches
---	Collimations assurance	See note 1	6.5	S3	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 (see note 10)					
---	Life-test provisions	Group C (see note 8); light output = 1,800 ftL; t = 500 hours (min)	---	---	---	---
---	Life-test end points	Light output (see note 33); cathode current (see notes 12 and 13)	---	1,600	---	ftL
		Collimations assurance (see note 1)	---	---	---	---
	Periodic-check tests (see note 40)					
---	Interaction factor	See note 29	---	---	1×10^{-5}	in./Vdc
1101	Secureness of base, cap, or insert	See note 8; immersion not required	---	---	---	---
---	Magnetic shielding	See note 32	---	---	---	---
1331	Direct-interelectrode capacitance	Cathode to all; writing gun	C	---	8	pF
		Grid 1 to all; writing gun	C	---	15	pF
		D1 to D2	C	---	3	pF
		D3 to D4	C	---	2	pF
		D1 to all	C	---	15	pF
		D2 to all	C	---	15	pF
		D3 to all	C	---	15	pF
		D4 to all	C	---	15	pF
		Suppressor grid to all	C	---	350	pF
---	Low-temperature operation	See note 37	---	---	---	---
---	Low pressure - high temperature	See note 11	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Periodic-check tests (see note 40) -Continued					
---	Vibration	See note 23	---	---	1	mm
---	Temperature cycling	See notes 8 and 30	---	5	---	Cycles
1002	Barometric pressure, reduced		---	---	---	---

NOTES:

1. Collimations assurance. The tube shall be operated at the conditions specified for the writing uniformity test in note 27. The voltages applied to the tube shall be varied one at a time as required by the following steps. At the voltage specified in each step, the writing uniformity of the tube shall be evaluated by the procedure in note 27. The writing uniformity at each step shall be within the limit specified in the writing uniformity test. At the end of each step, the voltage which was changed shall be returned to the condition specified in note 27. Erase pulse amplitude and noise input may be adjusted for each step to obtain the erase time and meter M1 reading required for the writing uniformity test, but no other voltages shall be changed.

   Step:

   1. Raise viewing gun heater voltage to 6.93 volts.
   2. Reduce viewing gun heater voltage to 5.67 volts.
   3. Replace the circuit of (test condition for collimated section-negative input) with a low-impedance supply of 260 volts at the positive input.
   4. Replace the circuit of (same as step 3) with a low-impedance supply of -100 volts at the negative input.
   5. Replace the circuit of (test condition for factory collimated section-positive input) with a low-impedance supply of 260 volts at the positive input.
   6. Replace the circuit of (same as step 5) with a low-impedance supply of 270 volts at the positive input.

2. The peak voltage between accelerating electrode (grid No. 4) and any deflecting electrode is ed.
3. The peak voltage between grid No. 1 and cathode is ec1.
4. Series current-limiting resistor to be used externally in screen circuit.
5. The writing gun grid No. 1 (control grid) shall not be operated more positive from cutoff than is necessary to write the display to the required brightness for a given scan rate and erase condition. In no case shall it be operated positive with respect to the writing gun cathode.
6. Total external circuit resistance in series with the backing electrode is Rbe.
7. Not required for qualification approval of the tube.
8. Operable reject tubes from current lot may be used for these tests provided the outcome of the tests shall not be affected by the defect causing the rejection. The original defect shall be disregarded when determining whether the tube under test passes or fails these tests.

   Tubes for life test may be rejects for mechanical defects only.

9. After method 1136 (rough-handling test), there shall be no damage to the external parts of the tube under test, and the tube under test shall meet all requirements of quality conformance inspection, parts 1 and 2, except dimensions and pressure.

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NOTES: -Continued

10. From each group of 50 tubes produced (or from each monthly production lot, if the lot exceeds 50 tubes) select one tube to be evaluated against these tests. One tube may be subjected in sequence to all tests or a separate tube from the same production lot may be selected for each test if concurrent testing is convenient. If this initial sample passes each required test, the group of 50 tubes (or the monthly lot) may be shipped. If a test is failed, two additional samples shall be subjected to that test and pass the test to permit release of the group which they represent. If either or both of these two samples fail, 100-percent product evaluation is required until such time as 10 successive tubes pass.

11. The tube shall be conditioned for a period of not less than 1 hour in a chamber having an ambient air temperature of +100° C. At the end of 1 hour, the tube shall be operated with normal voltages applied and at a pressure equivalent of 32,000 feet. Following this, the chamber temperature shall be reduced to +53° C. At the end of 1 hour, the tube shall be operated with normal voltages applied and a pressure equivalent of 60,000 feet altitude. Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. The criteria of satisfactory operation for a period of 15 minutes shall be the presence of a clearly defined circular trace display (absence of corona and arc-over).

12. Radar — Writing from visual extinction with no erase pulse applied, write a single frame of pattern with sweep speed of 36,000 in./sec, with pattern size adjusted so that lines are just merged (see method 5221). With Ec_1 voltage (not to exceed 50 volts above cutoff) adjust for brightness of at least 90 percent of saturation, measured within 3 seconds after cessation of writing. Cathode current shall not exceed 1,000 μAdc.

13. TV — Writing from visual extinction with no erase pulse applied, write a single frame at standard TV rates with a pattern size of 3.5 inches x 3.5 inches. With Ec_1 voltage (not to exceed 50 volts above cutoff) adjust for brightness of at least 50 percent of saturation, measured within 3 seconds after cessation of writing. Cathode current shall not exceed 1,500 μAdc.

14. With the writing guns cutoff under test conditions, operate viewing gun under the following conditions for a minimum of 5 minutes:

Ef:	6.3 volts	Ebe:	-20 Vdc
Ec1:	-100 Vdc	E_{supp}:	+150 Vdc
Ec2:	175 Vdc	Escn:	11,000 Vdc
Ec4:	350 Vdc	Ed:	400 Vdc

At the completion of the specified time, Escn shall be increased to 14,000 Vdc for 60 seconds. There shall be no recurrent breakdown between elements.

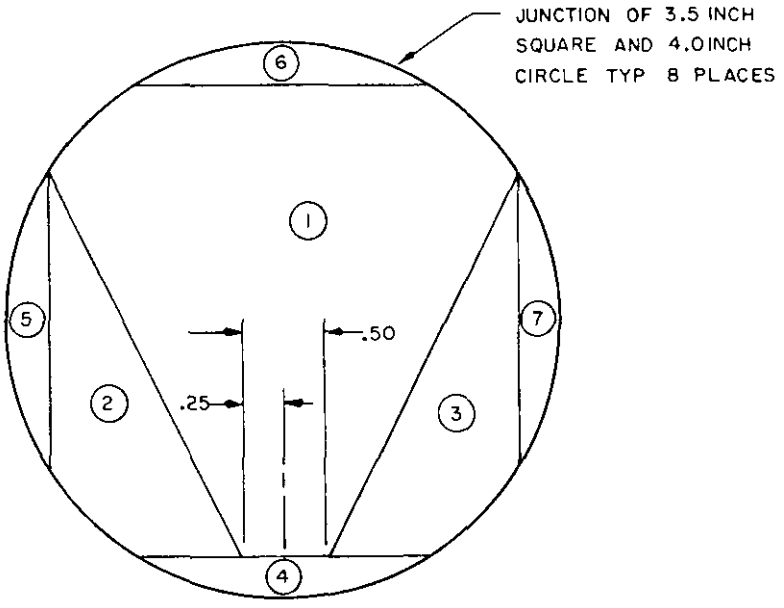
15. Write the viewing screen to saturation; increase the erase pulse amplitude to just effect visual cut-off; remove the erase pulse; no area shall reach 10 percent of saturation within 15 seconds as measured with a Spectra Brightness Spot Meter (Photo Research Corporation, or equivalent) at a distance of 5 feet (0.5-inch viewing diameter). Incident light on the area under test shall be less than 1 percent of saturated light output.

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16. Dark spots. Dark spots are defined as areas of the screen which do not brighten during writing or which require unusual processes to effect brightening; areas that lack persistence or whose persistence is obviously less than that of surrounding areas to an observer and includes the more easily recognized dead spots, burned spots and areas damaged by arc-overs, etc. The maximum number of dark spots is as follows, refer to figure (blemish zones):

	Number of spots in zone	
Maximum diameter of spot	*Zone 1*	*Zones 2 and 3*
0.015 inch to 0.040 inch	2	4 total

with a total of 5 in zone 1, 2, and 3. Dark spots permissible in zones 4, 5, 6, and 7 but shall not influence zones 1, 2, or 3. Dark spots less than 0.015 inch permissible provided their minimum separation is 0.3 inch.



BLEMISH ZONES

17. Light spots.

(a) Adjust the persistence of the tube to 3.5 seconds by adjusting the amplitude of the erase pulse. The erase pulse is to be a positive going rectangular pulse. (See note 44.)

(b) Prime the tube. Priming is defined as momentarily switching the backing electrode from its normal circuit to the collector and then back to its normal circuit.

(c) Observe all bright spots 3.5 ±0.5 seconds after returning the backing electrode to its normal circuit. Note their location and wait 30 seconds.

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NOTES: -Continued

17. -Continued

(d) Spots which do not erase in 30 seconds shall be defined as brilliant spots.

(e) Spots which erase in less than 30 seconds shall be defined as bright spots.

		Zone 1	Zones 2 and 3
Brilliant	1 any size	None	None
Bright	Greater than 0.030	None	None
Bright	Equal to or less than 0.030	2 0.5-inch (min) separation	4 total 0.25-inch (min) separation
		With a total of 5 in zones 1, 2, and 3	

18. Erasure shall be accomplished by applying rectangular pulses to the backing electrode (see note 44). Incident light on the area under test shall be less than 1 percent of saturated light output. Write the useful screen area to saturated brightness. Apply erase pulses with duty cycle of 20 μs on and 6,500 μs off. Persistence of 3.5 seconds shall be obtained when the erase pulse amplitude is varied in the range 2 to 10 volts peak-to-peak. The difference between the amplitude of an erase pulse required to brighten any area of the unwritten screen and the amplitude of an erase pulse required to completely illuminate the screen shall not exceed 20 percent of cutoff.

19. This test is applied with an approximately 95 V negative-going pulse applied to the suppressor grid. The pulse shall be of approximately the same pulse width as and in sync with the erase pulse. The specific amplitude and rise and fall time of this pulse shall be adequate to suppress the unwritten background light level to less than 5-foot lamberts.

20. The average voltage of each pair of deflection plates shall be equal to E_{c4}.

21. To measure direct view storage tube (DVST) resolution in terms of shrinking raster lines/inch at the same time that the light output is 1,000 ftL, proceed as follows:

Step 1.

(a) Adjust the display persistence to 3.5 ±0.5 seconds. Display a 45 to 80 line raster pattern on the DVST by applying a 60 Hz sawtooth voltage to the vertical axis deflection plates (D_3-D_4) and a 2,700 to 4,800 Hz sawtooth voltage to the horizontal axis deflection plates (D_1-D_2) of the writing gun under test. (Note: The selection of a specific number of lines will be determined by the size of the area measured and by the type of light meter being used.) Set the raster trace length (D_1-D_2) at 3-1/2 inches and center the raster on the tube face.

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21. -Continued

(b) Increase the low-frequency deflection amplitude (D_3-D_4) until the horizontal lines in the raster are individually visible. Now compress the raster by decreasing the low-frequency deflection amplitude (D_3-D_4) until the horizontal lines just merge together at the center of the raster and the individual line structure at that point disappears.

(c) An eye-corrected photometer whose measuring aperture is smaller than the height of the compressed raster shall be used to measure the light output. By adjustment of the control grid voltage of the writing gun under test, set the display light output to 1,000 ftL at the center of the compressed (just merged) raster. It may be necessary to further compress or expand the raster (decrease or increase the D_3-D_4 amplitude) in order to get it properly merged at this light level. The raster is properly merged when the line structure of the horizontal lines at the center of the raster just disappears as the D_3-D_4 amplitude is being decreased. Caution: Overmerging the lines may result in a condition where the line structure brightness reverses, giving brighter-than-normal line structure in the areas which formerly were spaces between lines. This condition is to be avoided as it introduces a significant error into the measurement.

(d) The resolution is calculated by dividing the number of horizontal lines visible in the raster by the height at the center of the properly merged raster. The resolution so calculated shall be no less than the value specified.

(e) De-center the raster in a vertical (D_3-D_4) direction and repeat steps 1 (a) through 1 (d) 1-1/2 inches above and 1-1/2 inches below the center of the tube face. The resolution at this point shall be no less than the value specified.

Step 2.

(a) Interchange the scanning axes on the writing gun just tested so that the 60 Hz sawtooth voltage is on the horizontal deflection plates (D_1-D_2) and the 2,700 to 4,800 Hz sawtooth voltage is on the vertical deflection plates (D_3-D_4). This will produce a 45 to 80 line vertical raster. Set the trace length (D_3-D_4) at 3-1/2 inches and center the raster on the tube face.

(b) Increase the low-frequency deflection amplitude (D_1-D_2) until the vertical lines in the raster are individually visible. Now compress the raster by decreasing the low-frequency deflection amplitude (D_1-D_2) until the vertical lines just merge together and the individual line structure disappears.

(c) Using the photometer described in step 1 (c), above, measure the light output. By adjustment of the control-grid voltage of the writing gun under test, set the display light output to 1,000 ftL at the center of the compressed (just merged) raster. It may be necessary to further compress or expand the raster (decrease or increase the D_1-D_2 amplitude) in order to obtain a properly merged raster at this light level. The raster is properly merged when the individual structure of the vertical lines just disappears.

(d) The resolution is calculated by dividing the number of vertical lines in the raster by the width of the properly merged raster. The resolution so calculated shall be no less than the value specified.

(e) De-center the raster horizontally and repeat the procedure 1-1/2 inches to the left and 1-1/2 inches to the right of the center of the tube face. The resolution at the point selected shall be no less than the value specified.

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21. -Continued

Step 3.

Repeat all procedures, step 1 (a) through step 2 (e) for the other writing gun.

22. Adjust erasure to a convenient value. Do not drive the writing gun grid No. 1 (control grid) any harder than necessary to achieve a useful spot brightness. Overdrive on a stationary or slow moving spot will cause damage to the DVST.

Connect each deflecting electrode through a separate 100 K-ohm resistor to G_4 of the writing gun. With means for short-circuiting each resistor, measure the spot displacement when each resistor is short-circuited in turn.

23. The tube shall be suitably mounted on the vibration table by large base diameter and zone C (see figure 4) and tested as follows:

(a) With the tube mounted and with a circular trace positioned on the tube face and operating at prevailing room temperature, the tube shall be vibrated along the X-X axis with the frequency varying between 10 and 500 Hz at an amplitude as indicated by vibration specification curve (see figure 4) as monitored at a point immediately adjacent to one of the mounting areas. The rate of frequency change shall be approximately uniform, and shall be such that a complete cycle, for example, 10 to 500 to 10 Hz, will consume approximately 15 minutes. The test shall continue for 90 minutes. During this time, all resonance points shall be noted and the increase in linewidth of the circular trace shall be within the limits specified herein. To accurately determine resonance points, the frequency shall be varied slowly through the range around each of the noted resonance points. A point of resonance is defined as that frequency at which a noticeable change in the written or stored information on the screen is visible. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits, or for other indications of tube malfunction or degration in performance.

(b) With the tube operating, the tube shall be vibrated for 30 minutes at the resonant frequency noted in step (a). Where more than one resonant frequency is noted, the test period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. Where resonant frequencies are not apparent within the specified frequency range, the tube shall be vibrated for a period of 1 hour at a frequency of 55 Hz. During this time, the increase in linewidth of the circular trace shall be within the limits specified herein. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits or for indications of tube malfunction or degradation in performance.

(c) Steps (a) and (b) shall be repeated with the tube mounted such that the vibration is along the Y-Y axis.

(d) Steps (a) and (b) shall be repeated with the tube mounted such that the vibration is along the Z-Z axis.

(e) Following vibration, retest tube. All requirements and test limits specified in quality conformance inspection, part 1 (except vibration) and part 2 of this specification shall be satisfied. In the event a reject tube has been used for this test, under the conditions outlined in note 8, the item which was cause for initial rejection may be disregarded.

NOTE: This test is classified as a destructive test, in the sense that tubes subjected to it shall not be shipped.

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24. Each gun: From cutoff to +50 volts above cutoff: Ec3 = 200 Vdc (max). Spot shall be pulsed to prevent damage to the tube.

25. The total movement of the ends of a 2.8-inch trace (centered with respect to the tube face), deflected horizontally or vertically 1.40 inches each side of the center of the tube face, shall not exceed 0.060 inch. The 1D2 trace shall be considered horizontal and the 3D4 trace shall be considered vertical.

    Alternate method for measuring pattern distortion. A rectangular raster shall not fall outside a square frame whose inside dimension is 2.74 inches and outside dimension is 2.86 inches and whose center coincides with the centerline of the tube.

26. The dc resistance between any element including the magnetic shield but excluding the heaters, backing electrode (Ec5), and suppressor (Ec6) and all other elements connected together measured at a potential of 250 Vdc shall be no less than specified. The dc resistance between Ec5 and Ec6 and all other elements connected together measured at a potential of 150 Vdc shall be no less than specified. Should the polarity of the measuring potential influence the resistive reading, reverse the polarity and the lower of the two readings shall apply. The heaters need not be energized during this test.

27. During the grid-drive requirements (writing gun) test, performance is judged by feeding a random noise signal of known amplitude to each of the two writing guns ("A" gun to be sampled per AQL 1.0 percent inspection level II) in succession and measuring the resulting display luminescence as a percentage of saturated light output. The writing uniformity test shall be performed immediately following and in conjunction with grid drive. The equipment to be used, the conditions of test, the calibration procedure, and the sequence of test operations are described below:

    1. Equipment.

        (a) Random noise generator (5 MHz scale) - General Radio Type 1390-B, or equivalent.

        (b) Sine wave generator (5,000 to 9,000 Hz) - Hewlett Packard Model 200 ABR, or equivalent. (Note: Sine wave shall be non-synchronous with vertical pulse repetition frequency (PRF)).

        (c) Video gating and mixing circuitry designed to accept, successively, the output from the random noise and sine wave generators. These signals shall be gated on only during the mutual period of the vertical scan and the pedestal. Negative-going portions of noise shall be rejected; i.e., clipped, and working into a 2 K ±5 percent load. Noise and pedestal should add. Do not clip calibrating sine wave. Response of the system up to and including the DVST write gun grid to be 2 MHz minimum at the 3 db point.

        (d) Appropriate scanning waveform, pedestal, and erase pulse generators. (See conditions of test, below.)

        (e) Noise calibrating circuitry in accordance with figure 3.

        (f) Light output measuring device in accordance with figure 3.

    2. Conditions of test.

        (a) Vertical scan height: 3.5 inches (centered).

        (b) Vertical scan time: 625 μs (linear sawtooth).

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27. -Continued

(c) Vertical PRF: 500 pps.

(d) Horizontal scan width: 3.5 inches (centered).

(e) Horizontal scan rate: 3.5 in./sec (linear triangular).

(f) Mean deflecting electrode potential: 100 Vdc ±3 percent.

(g) Writing gun G1 input: Output waveform from noise calibrating circuitry, clamped to G1 bias level.

(h) Noise calibrating circuitry video inputs to point A.

a. Pedestal only (625 μs rectangular pulse coincident with vertical scan period).

b. Pedestal plus gated sine wave.

c. Pedestal plus gated and clipped noise.

(i) Noise calibrating circuitry plus input to point B; pedestal only.

(j) Erase pulse; positive going rectangular pulse (20 μs on and 6,500 μs off.

(k) Other tube voltages in accordance with test conditions.

3. Calibration procedure.

(a) With pedestal only applied to point A, adjust pedestal amplitude for 10 volt peak-to-peak reading at point Eo of figure 3. No further adjustment permissible.

(b) With 12 volt peak-to-peak pedestal at input B adjust R7 of noise calibrating circuitry for null reading on meter M1. No further adjustment permissible.

(c) Apply sine wave to pedestal; i.e., pedestal plus gated sine wave shall be applied to point A. Adjust amplitude of gated sine wave to 23 ±0.5 volts peak-to-peak as indicated by a calibrated oscilloscope at point Eo of figure 3. Read and record meter M1. Remove gated sine wave from pedestal.

(d) Cover opening to photoelectric cell V3, close switch S1, and adjust potentiometer R15 for null reading of meter M2.

(e) Write display to saturated brightness by reducing erase pulse amplitude to zero, adjusting writing bias so that pedestal writes fully a "stored" blank raster, and then adjusting write bias to cutoff.

(f) Place photoelectric cell V3 against the faceplate of the DVST and after steady reading is obtained, read and record meter M2.

NOTE: Avoid saturation of the measuring device by the addition of sufficient neutral density filters in front of the photoelectric cell. A reading of approximately 4.5 volts indicates saturation. Sufficient filtering to obtain an average saturation reading between 3 and 4 volts shall be required.

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NOTES: -Continued

27. -Continued

4. Test sequence.

(a) Apply erase pulse as in 2 (j). Amplitude should be between 2 and 10 volts, peak-to-peak.

(b) Adjust bias of writing gun G1 under test, with pedestal only, until meter M2 reads the same as step 3 (f).

(c) Turn off writing gun when sweep is in center and note the time required for saturated light output to be erased to background brightness (complete erasure over entire face of tube).

(d) Repeat steps 4 (b) and 4 (c) adjusting erasure pulse amplitude to obtain 3.5 ±0.5 second persistence.

(e) Apply pedestal only to point A. Adjust bias of writing gun G1 under test just to visual cutoff.

(f) Apply pedestal plus gated and clipped noise to point A (figure 3). Adjust amplitude of noise until meter M1 reads the same as in (3) (c). (If noise peaks exceed 32 V, CR1, CR2, CR3, and CR4 in figure 3 may be defective.)

(g) Place photoelectric cell against faceplate of DVST. Close switch S1 momentarily. Release switch S1 and allow sufficient time for integration of noise to reach equilibrium. Scan the face of DVST with photocell to find any record the minimum and maximum readings of M2.

(h) The minimum reading in step 4 (g) shall be not less than 20 percent of the meter reading in step 3 (f).

(i) If the minimum reading observed in step 4 (g) is less than 50 percent of the meter reading in step 3 (f), the ratio of the maximum to minimum readings in 4 (g) shall not exceed the value specified for writing uniformity.

(j) If the minimum reading in step 4 (g) is greater than 50 percent of the meter reading in step 3 (f), reduce the calibrated noise input to obtain a value which is 40 percent of the meter reading in step 3 (f). Again scan the face of the DVST for minimum and maximum values. The ratio of maximum to minimum readings shall not exceed the value specified for writing uniformity.

(k) Using pedestal only, repeat step (c) to insure that persistence is 3.5 ±0.5 second.

28. As a part of qualification, two tubes shall be operated in a completed radar system for a period of 6 hours in each, and shall demonstrate satisfactory performance in the equipment.

29. The deflection per unit volt of one beam at any position of the beam within its useful scan, when balanced voltage of at least ±300 Vdc, applied to the 1D2 deflection electrodes of the other electron gun shall be not greater than the value specified herein.

30. All heaters shall be energized with nominal heater voltage during that portion of each cycle beginning at low-temperature equilibrium and ending at high-temperature equilibrium. Use MIL-STD-202, method 107, condition B, except temperature in step 3 shall be 100° ±3° C.

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NOTES: -Continued

31. This is a 100 percent test and shall be applied to all tubes. With the tube mounted as specified in note 23, with the vertical reference trace positioned vertically, and operating at prevailing room temperature, the tube shall be vibrated along the Y-Y axis with the frequency varying between 10 and 500 Hz at an amplitude as indicated by the vibration specification curve as monitored etc. (same as in note 23). The rate of frequency change shall be approximately uniform, and shall be such that a complete cycle (10 to 500 to 10 Hz) will consume approximately 10 minutes. The tube shall be vibrated for one cycle. During this period, maximum particle activity shall be observed and recorded in the lower and upper frequency ranges (10 to 100 Hz and 100 to 500 Hz). Select a single fixed frequency in each of the two ranges where particle activity was observed and vibrate the DVST for an additional 5 minutes at each of the frequencies. Tubes which display blemishes on the screen, show arcs or flashing, or develop permanent or temporary shorts, shall be rejected. Following vibration, retest the tube. All initial requirements and test limits shall apply and the tube shall be within initial limits.

32. The shielding shall be such that 10 gauss field shall not deflect an undeflected spot more than 1 mm and shall not otherwise affect the performance of the storage tube. The magnetic shield shall be secured to the tube at least to the full length of zones A, B, and C, as shown on figure 1 with an encapsulating material, electrically nonconducting and non-nutrient. The shield shall be capable of withstanding a magnetic field when tested by the following or equivalent procedure:

    (a) A suitable pair of air-core coils to produce the magnetic field for testing has 375 ampere-turns each, a mean cross-section of 10.5 by 2.75 inches perpendicular to the field and a separation of 8.5 inches. The center of the coils shall be approximately centrally located about the tube.

    (b) With the tube operating as for the spot position test, measure spot motion when the coil current is interrupted. The spot motion under these conditions shall not exceed 1.0 mm. Note any effects of the magnetic field on the viewing beam.

33. At the end of 500 hours, the tube shall be retested and shall meet initial requirements of quality conformance inspection, part 1, except light output and vibration.

34. Single pulse erase time. With display at equilibrium (so defined in note 36) apply a single rectangular erase pulse equal to amplitude required to produce a 3.5 ±0.5 second persistance (note 18) plus 2 volts. The single pulse width should equal 8 ±1 milliseconds. Decrease amplitude of pulse slightly and repeat above until a minimum pulse amplitude is obtained which gives the lowest value of erase time. Working at this amplitude, decrease pulse width to lowest value which will still produce the condition of visual cutoff.

    The just-visual cutoff erase time (i.e., the width of the single pulse) measured at the condition of minimum pulse width and minimum amplitude shall be equal to or less than the limit stated.

35. Prior to the performance of any quality conformance inspection tests, the tube shall be operated for a minimum of 48 hours.

36. Measured at equilibrium with Spectra Spot Brightness meter, or equivalent. Light output is measured over any half inch, or greater, area any place within the useful viewing diameter. Equilibrium is that level of brightness obtained several seconds after cessation of writing with no erase pulse applied and where the entire screen area has been written to the maximum obtainable brightness (saturation brightness).

37. Tubes shall be cold soaked for a period of 48 hours at -65° C. At the end of this period and while still at the low temperature, all heater voltages shall be applied for a period of 15 minutes. Tubes shall then be removed from freezer and tested for storage time.

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NOTES: -Continued

38. The following guide lines shall apply:

    (a) Tubes which exhibit momentary bursts of particle activity on the application of voltage and in which particles show no clinging action to cause blemishes, shall be considered acceptable provided the tubes pass vibration test, note 31.

    (b) Tubes which exhibit particle activity on the application of voltages or under vibration to a degree which causes blemishes, arcing, repetitive flashing, or other disturbance not normal to storage tube operation, shall not be considered acceptable.

39. After 60 seconds of heater-cathode warmup time, application of normal tube voltage will produce a collimated 4-inch (min) display.

40. Shipment of tubes prior to completion of quality conformance inspection, part 2 and part 3 (periodic-check tests) is permitted provided quality conformance, part 2 and part 3 (periodic-check tests) have been successfully completed on the preceding month's lot.

41. The factory collimated section includes all viewing gun electrodes except screen, suppressor grid, backing electrode, and heater. The reference input should be grounded.

42. Resolution, Gray Shades, and uniformity in the TV mode are measured differently from the corresponding parameters when the tube is used in radar mode. In measuring all three parameters in the TV mode the following setup conditions are required:

    (a) Supply horizontal and vertical sweep at standard EIA rates of 525 scanning lines, 60 fields per second, 30 frames per second, interlaced 2:1.

    (b) Adjust display raster size to 3.5 x 3.5 inches, centered on screen.

    (c) Apply erase train as described in note 44.

    Uniformity. Adjust writing gun control grid (Ec_1) until the brightest area on the raster reaches 750 ftL. Without further changing Ec_1, measure and record the brightness both at this point and at the point on the raster of least brightness. The ratio of highest to lowest brightness is the television writing uniformity and shall be no higher than the value specified.

    Gray shades. A "stair-step" generator capable of producing a signal equivalent to the RETMA chart gray scales is to be used. The signal consisting of 9 to 10 vertical bars of equal width and uniform steps from black (zero volts video) to white is to be applied to Ec_1. This signal amplitude shall be adjusted for optimum gray scale presentation. The number of gray scales discernable on the display is to be no less than the number specified.

    Resolution. The same scan rates are used as on the uniformity and halftone measurements; however, the writing beam is modulated at the control grid from either a sine or square-wave generator whose output amplitude is constant with frequency to at least 6.5 MHz, and which can be keyed with the horizontal sweep or sync so as to start a cycle at the start of each horizontal line. The measured resolution shall be as specified. Procedure is as follows:

    (a) Apply TV rate erase train as in uniformity and Gray Scale measurements.

    (b) Scanning at EIA rates, adjust bias of write gun to just cutoff.

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NOTES: -Continued

42. -Continued

(c) Adjust generator to 1.0 MHz and feed to control grid of writing gun under test. Adjust generator output (or video amplifier, if one is used) to produce a pattern of alternate black and white vertical lines on which the white lines achieve a brightness no less than the figure specified. Increase frequency to 4.0 MHz, making certain that the signal to the control grid maintains the same amplitude as that at 1.0 MHz. The visual amplitude response at 4.0 MHz shall be no less than 10 percent peak-to-peak of the amplitude response at 1.0 MHz.

43. The test is performed with no pulse train applied to the suppressor grid. The suppressor grid potential during the test is set to its nominal dc value.

44. The required values of erase pulse width and pulse repetition frequency (PRF) are determined by the display persistence desired. These should be selected on the basis of required system performance.

Operation at standard TV rates (525 scanning lines; 30 Hz frame rate, interlaced 2:1) requires that complete erasure occur each frame in order to avoid smearing of moving objects on successive frames. To achieve complete erasure and to avoid interference between the erase train and the scanning pattern at TV rates, erasure is best done during the writing beam flyback interval (both horizontal and vertical). Horizontal blanking time is approximately 10 μs per line and vertical is approximately 1,160 μs per field in the standard TV composite video signal. Total blanking time per frame, then, is approximately 7.2 milliseconds, or about 22 percent of the total frame time.

In the case of both radar and TV displays, the erase pulse amplitude for a desired persistence should be equal to storage surface visual cutoff if maximum writing sensitivity (target detectability) is to be maintained. Higher amplitudes reduce sensitivity through over-erasure; lower amplitudes permit background buildup through insufficient erasure. The change in storage surface visual cutoff for a range of persistence from 0.03 to 3 seconds is approximately 2 volts in tube type 8832. The corresponding range of erasure pulse duty cycles is from approximately 20 percent to 0.2 percent.

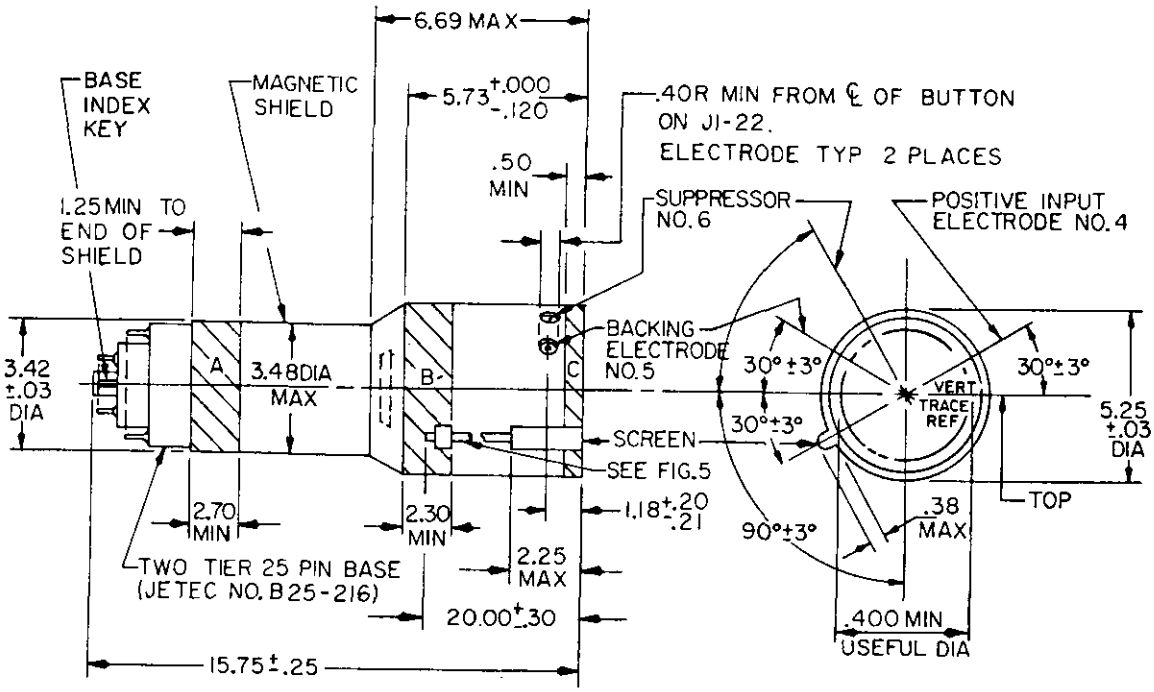
Custodian:
Air Force - 85

Review activities:
Air Force - 11, 17,

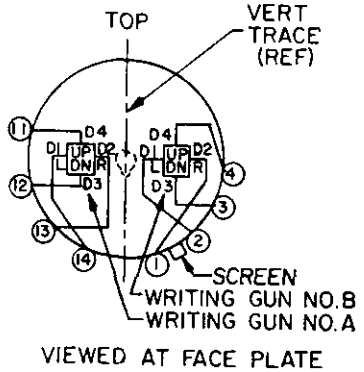
Preparing activity:
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(Project 5960-F765)

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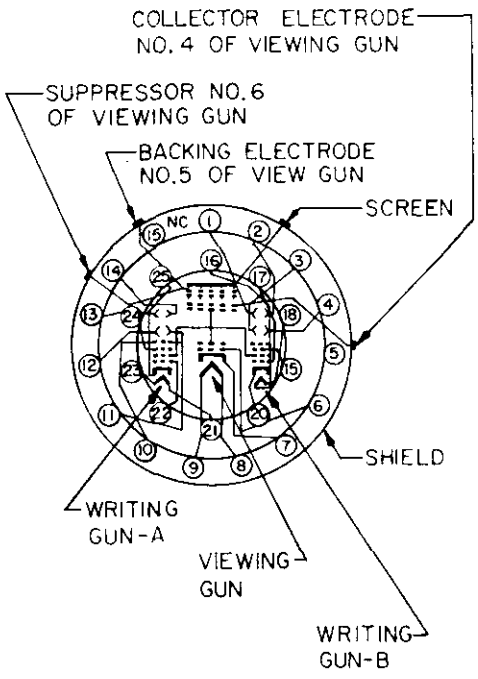
INCHES	MM	INCHES	MM
.03	.76	1.25	31.75
.120	3.05	2.25	57.15
.125	3.18	2.30	58.42
.20	5.08	2.76	70.10
.21	5.33	3.42	86.87
.25	6.35	3.48	88.39
.30	7.62	5.25	133.35
.38	9.65	5.73	145.54
.400	10.16	6.69	169.93
.50	12.70	15.75	400.05
1.18	29.97	20.00	508.00



NOTE:
Dimensions are in inches.

FIGURE 1. Outline drawing of electron tube type 8832.

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Pin No. 1 Deflecting electrode D2 of writing gun B.
Pin No. 2 Deflecting electrode D1 of writing gun B.
Pin No. 3 Deflecting electrode D3 of writing gun B.
Pin No. 4 Deflecting electrode D4 of writing gun B.
Pin No. 5 Shield
Pin No. 6 Negative input
Pin No. 7 Reference input
Pin No. 8 Heater of the viewing gun
Pin No. 9 Heater of the viewing gun.
Pin No. 10 Accelerating electrodes (No. 2 and No. 4 of writing guns)
Pin No. 11 Deflecting electrode D4 of writing gun A.
Pin No. 12 Deflecting electrode D3 of writing gun A.
Pin No. 13 Deflecting electrode D2 of writing gun A.
Pin No. 14 Deflecting electrode D1 of writing gun A.
Pin No. 15 No connection
Pin No. 16 Heater of writing gun B.
Pin No. 17 Control grid (No. 1) of writing gun B.
Pin No. 18 Heater of writing gun B.
Pin No. 19 Focusing electrode (No. 3) of writing gun B.
Pin No. 20 Cathode of writing gun B.
Pin No. 21 Heater of writing gun A.
Pin No. 22 Control grid (No. 1) of writing gun A.
Pin No. 23 Heater of writing gun A.
Pin No. 24 Focusing electrode (No. 3) of writing gun A.
Pin No. 25 Cathode of writing gun A.

FIGURE 2. Pin connections.

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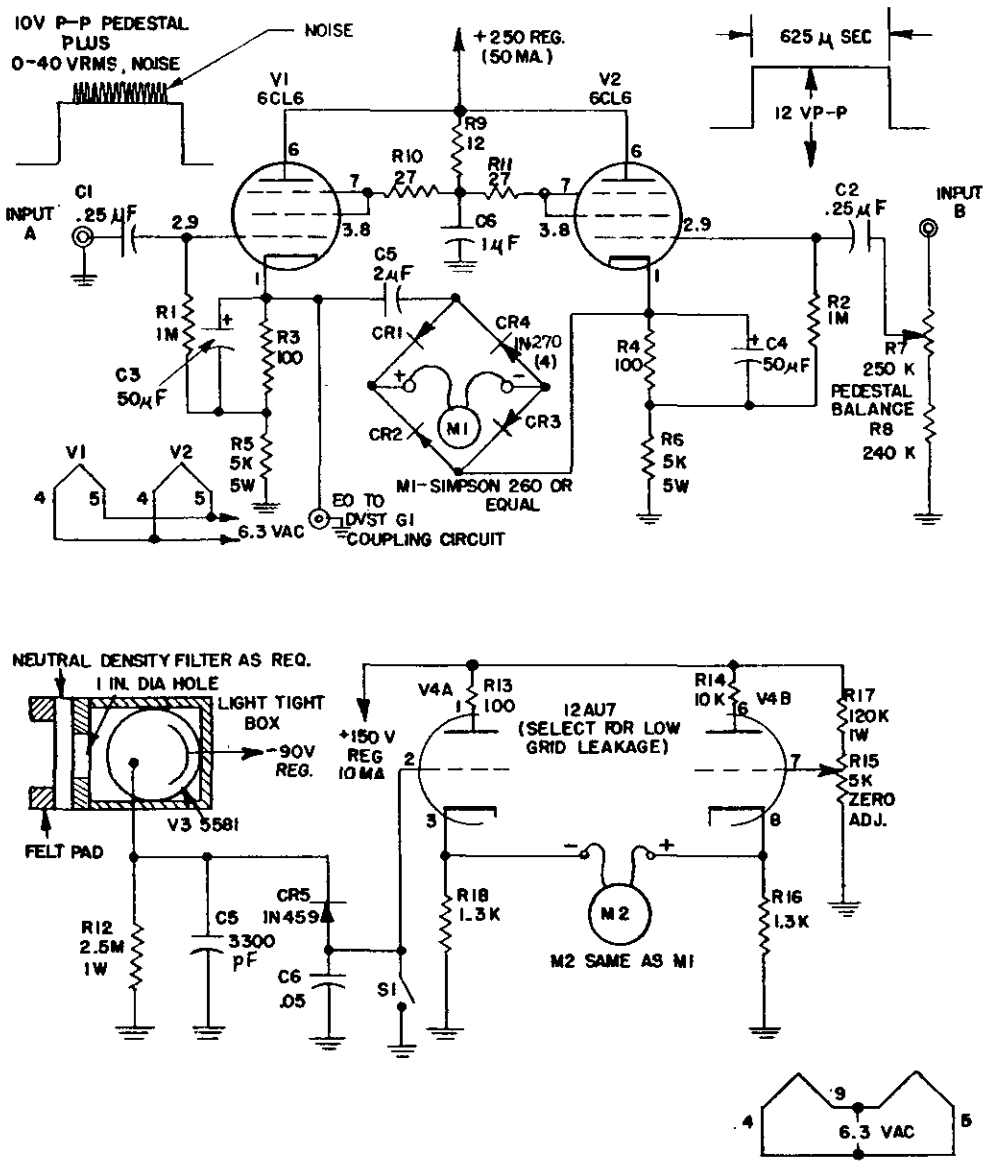


FIGURE 3. Grid drive test equipment.

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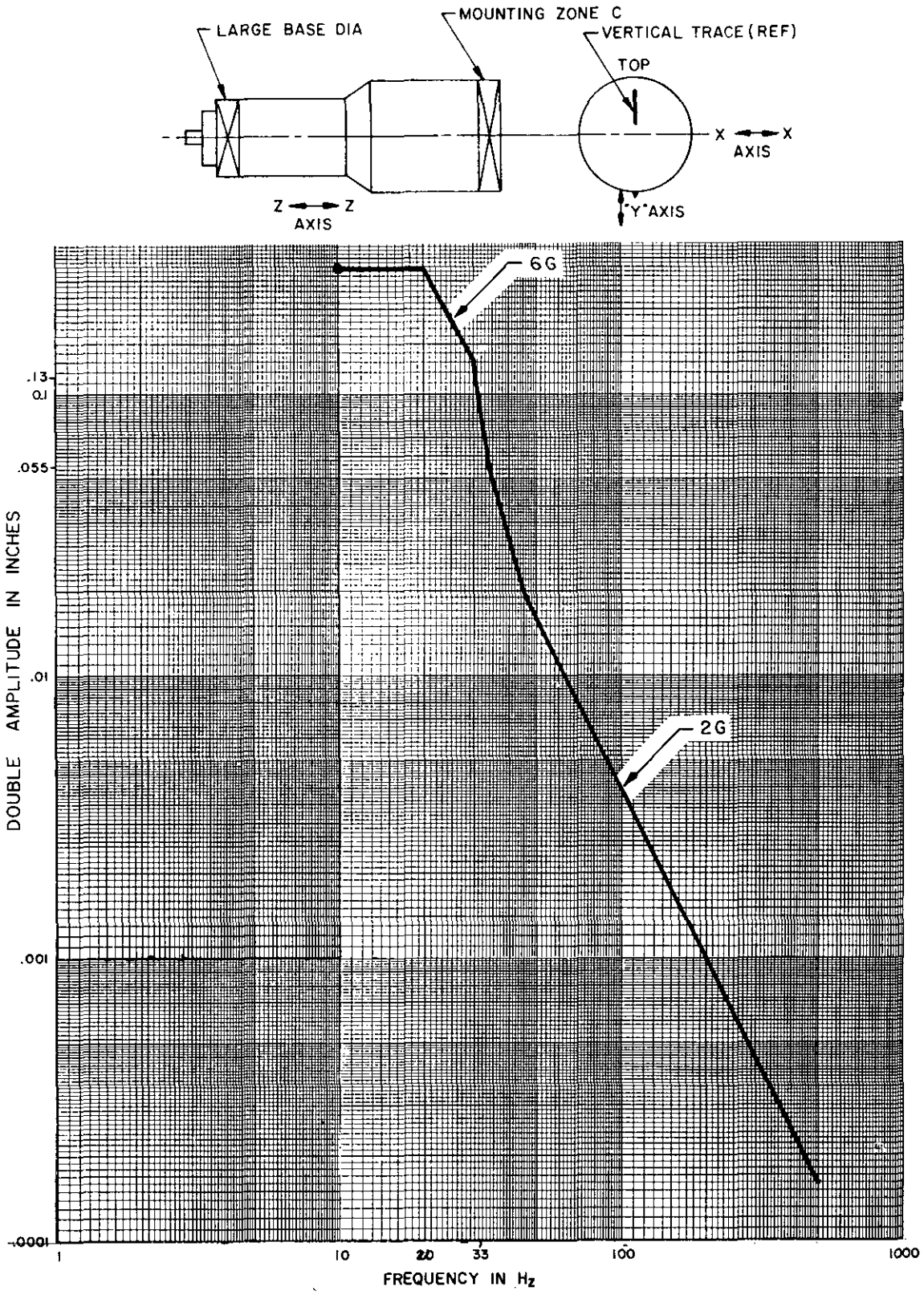
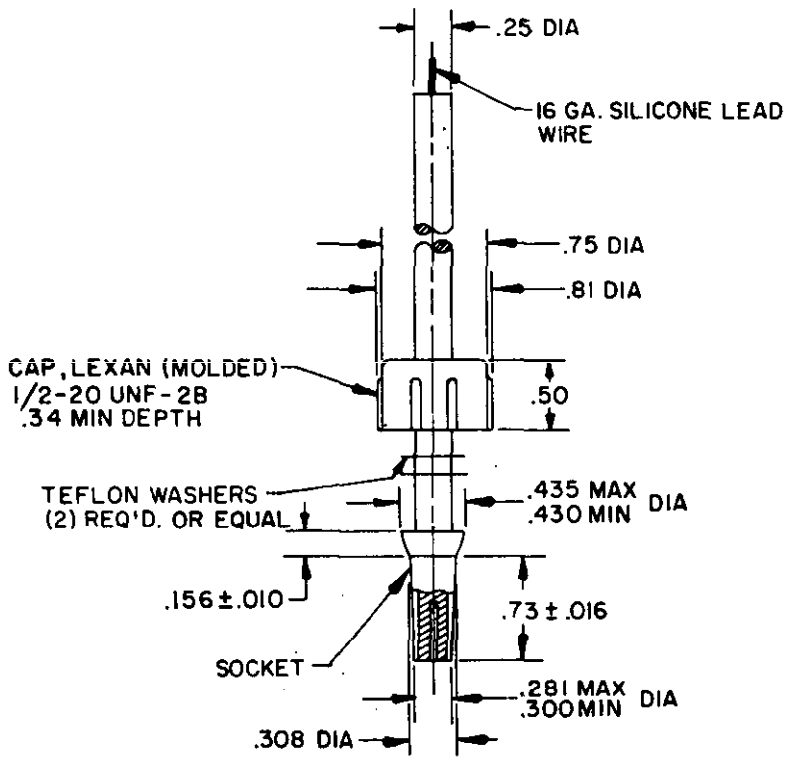


FIGURE 4. Vibration requirements.

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INCHES	MM	INCHES	MM
.010	.25	.34	8.64
.016	.41	.430	10.92
.156	3.96	.435	11.05
.25	6.35	.73	18.54
.281	7.14	.75	19.05
.300	7.62	.81	20.57
.308	7.82		

NOTE:
Dimensions are in inches

FIGURE 5. Screen connection.

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